MATHEMATICAL MELODIES

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BACKGROUND RESEARCH

MUSIC IS BUILT ON PATTERNS, AND PATTERNS ARE THE FOUNDATION OF MATHEMATICS. SEVERAL MATHEMATICAL IDEAS APPEAR NATURALLY IN MUSICAL STRUCTURE:

- RATIOS AND FREQUENCIES: NOTES SOUND GOOD TOGETHER WHEN THEIR SOUND WAVES FORM SIMPLE RATIOS. FOR EXAMPLE, OCTAVES HAS A FREQUENCY RATIO OF 2:1.
- SCALES AND INTERVALS: THE SPACING BETWEEN NOTES IN A SCALE FOLLOWS MATHEMATICAL RULES. THE EQUAL-TEMPERED SCALE DIVIDES AN OCTAVE INTO 12 EQUAL PARTS USING EXPONENTIAL FUNCTIONS.
- RHYTHM AND COUNTING: BEATS, MEASURES, AND TIME SIGNATURES ARE ALL BASED ON COUNTING AND FRACTIONS.

THIS SHOWS THAT MUSIC ALREADY FOLLOWS MATHEMATICAL RULES, WHICH SUPPORTS THE IDEA THAT MATH COULD ALSO CREATE MUSIC.

PROBLEM STATEMENT

CAN MATHEMATICAL CONCEPTS BE USED TO GENERATE MUSIC?

OUR GOAL

OUR GOAL IS TO BUILD A SOFTWARE TOOL THAT CREATES MUSIC BY MATCHING A MATH CONCEPT WITH A MUSICAL INSTRUMENT. THE GENERATED MUSIC IS THEN ANALYZED USING SPECIFIC METRICS TO SEE IF THE PAIRING WORKS WELL. ONCE WE KNOW WHICH MATH CONCEPT FITS BEST WITH WHICH INSTRUMENT, WE CAN USE THOSE COMBINATIONS TO CREATE MELODIOUS MUSIC.

HYPOTHESIS

MUSIC IS BASED ON PATTERNS, AND MATH CAN CREATE THOSE PATTERNS. IF WE IDENTIFY THE RIGHT MATH CONCEPT AND MATCH IT WITH THE RIGHT INSTRUMENT, WE CAN GENERATE MELODIOUS MUSIC.

TECHNOLOGY USED

VSCODE

STREAMLIT

MIDO PYTHON LIBRARY

MUSIC21 PYTHON LIBRARY

PANDAS PYTHON LIBRARY

PROCEDURE

1. RESEARCH PHASE: WE STUDIED BASIC MUSIC THEORY, MATHEMATICAL ALGORITHMS, AND PYTHON-BASED MUSIC LIBRARIES.
2. ALGORITHM DEVELOPMENT: WE WROTE PYTHON FUNCTIONS TO GENERATE FIBONACCI SEQUENCES AND BASIC MUSIC, SUCH AS MAJOR SCALES.
3. MUSIC GENERATION: WE USED THE MIDO LIBRARY TO MAKE MIDI FILES AND THE ABILITY TO ADJUST TEMPO, MELODY LENGTH, AND SCALE.
4. ANALYSIS DEVELOPMENT: WE RESEARCHED WHAT KINDS OF MUSICAL PROPERTIES FIT BEST WITH DIFFERENT CLASSES OF INSTRUMENTS AND DETERMINE HOW EACH MATH CONCEPT RELATES TO THESE PROPERTIES.
5. TESTING AND DEBUGGING: WE TESTED DIFFERENT OUTPUTS USING DIFFERENT SEQUENCES AND EVALUATE IF THE SOFTWARE CAN CORRECTLY DETERMINE IF THE MATH CONCEPT WORKS WITH THE INSTRUMENT.

MATHEMATICAL CONCEPTS

WE USED FOUR DIFFERENT MATH CONCEPTS:

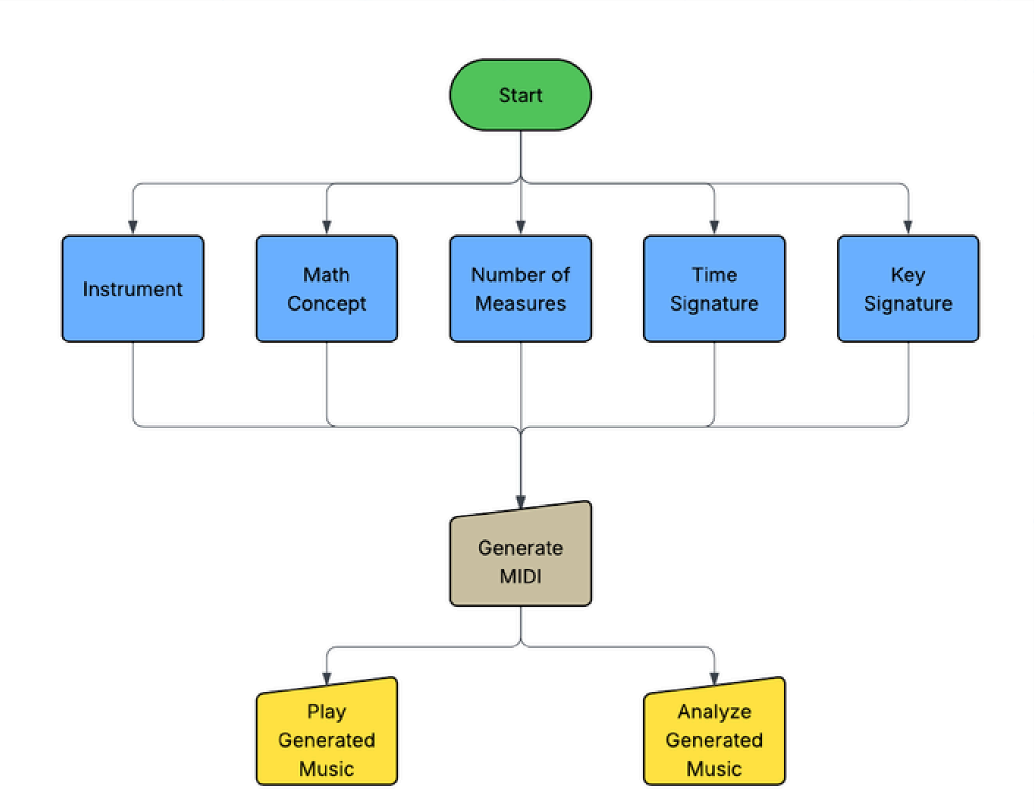
- FIBONACCI SEQUENCE - ALLOWS FOR NATURAL MUSICAL GROWTH
- PRIME NUMBERS - CREATES COMPLEX AND UNPREDICTABLE RHYTHMS
- MULTIPLES OF 2 - ALLOWS FOR NATURAL PROGRESSION AND AND RHYTHM
- MULTIPLES OF 5 - GIVES A SYNCOPATED FEEL TO THE MUSIC

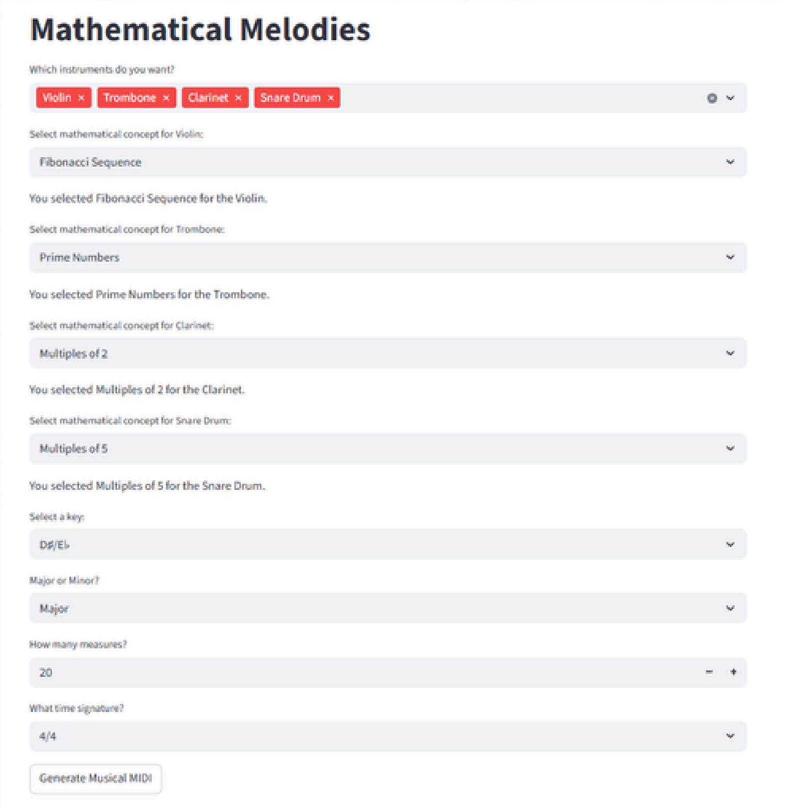
METRICS FOR ANALYSIS

WE HAVE FOUR DIFFERENT CATEGORIES OF ANALYSIS:

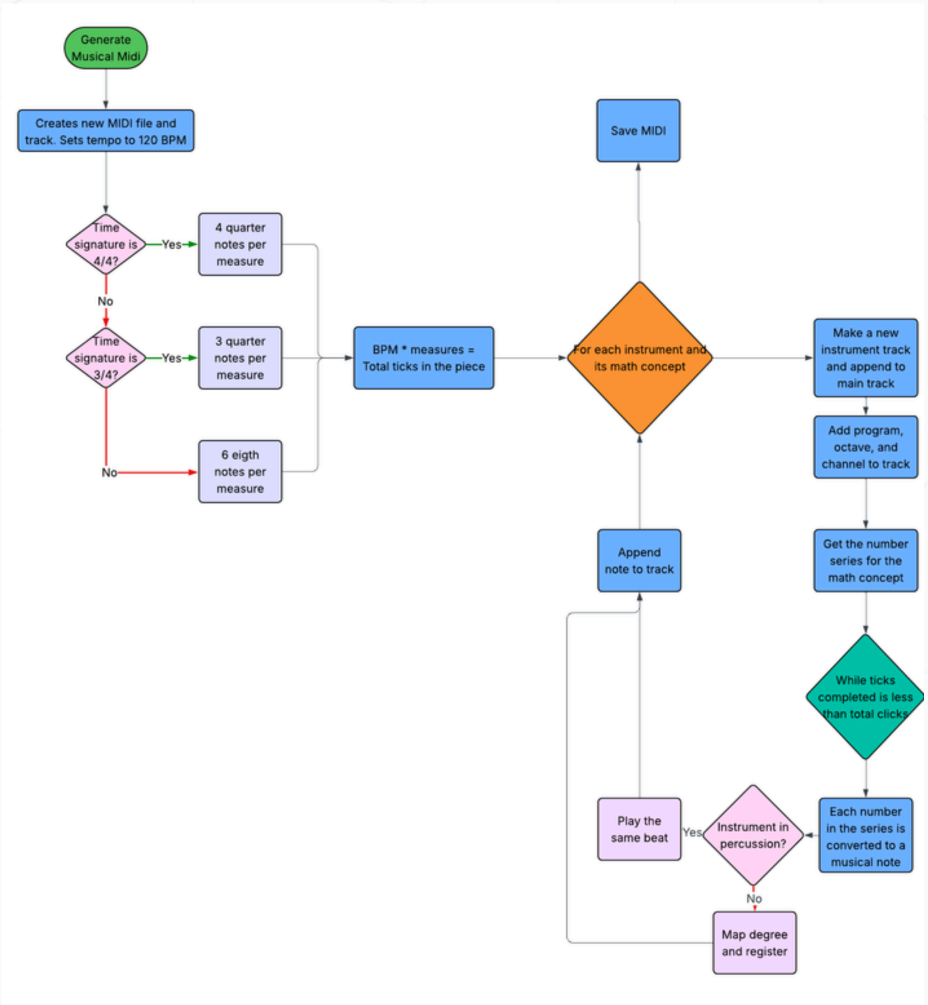
- INTERVAL DISTRIBUTION - THE AVERAGE JUMP IN CONSECUTIVE NOTES.
- RHYTHMIC DENSITY - THE AVERAGE AMOUNT OF NOTES PER BEAT.
- HARMONIC COMPLEXITY - THE AMOUNT OF VARIED CHORDS IN A BEAT.
- MELODIC CONTOUR - THE DIRECTION THE MUSIC IS GOING IN.

INPUT


Start
Instrument
Math Concept
Number of Measures
Time Signature
Key Signature
Generate MIDI
Play Generated Music
Analyze Generated Music

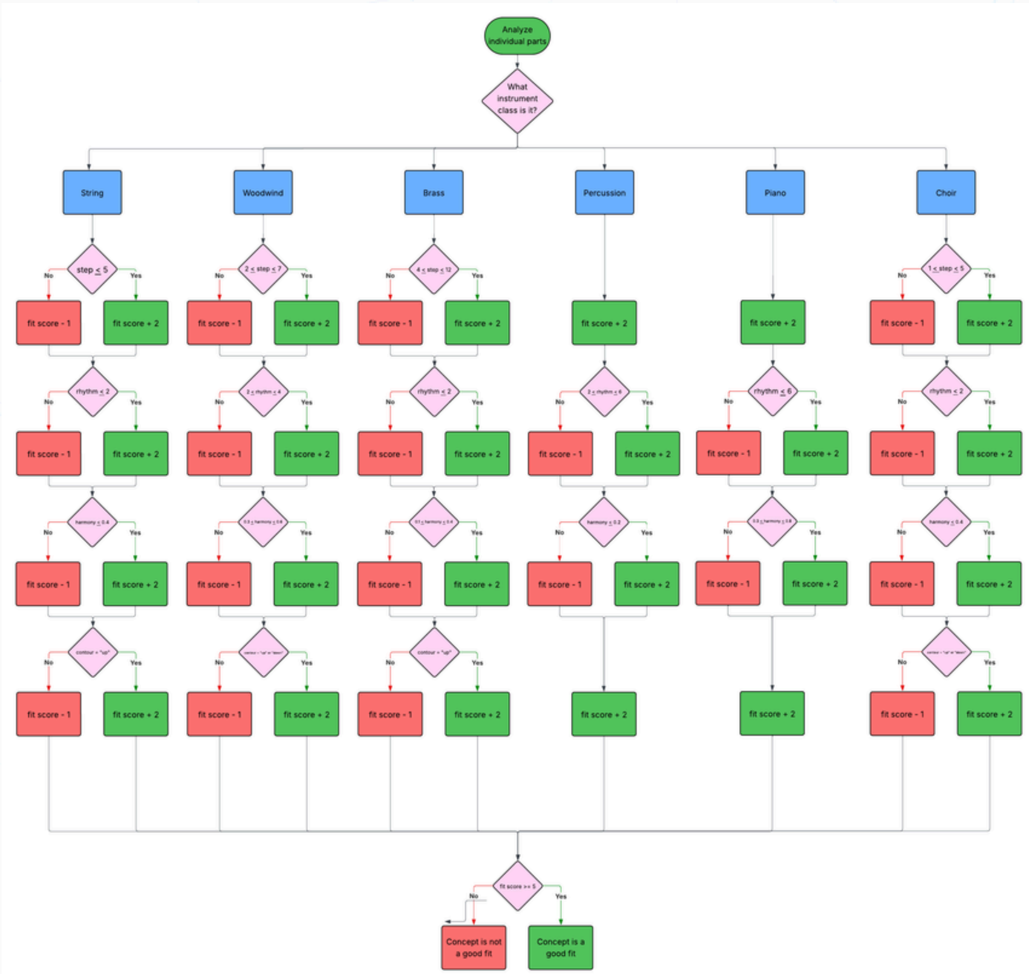

Mathematical Melodies
Which instruments do you want?
Violin ×
Trombone ×
Clarinet ×
Snare Drum ×
Select mathematical concept for Violin:
Fibonacci Sequence
You selected Fibonacci Sequence for the Violin.
Select mathematical concept for Trombone:
Prime Numbers
You selected Prime Numbers for the Trombone.
Select mathematical concept for Clarinet:
Multiples of 2
You selected Multiples of 2 for the Clarinet.
Select mathematical concept for Snare Drum:
Multiples of 5
You selected Multiples of 5 for the Snare Drum.
Select a key:
D♯/E♭
Major or Minor?
Major
How many measures?
20
What time signature?
4/4
Generate Musical MIDI

MUSIC GENERATION LOGIC



- THE SYSTEM STARTS BY CREATING A MIDI FILE WITH A DEFAULT TEMPO OF 120 BPM AND DETERMINING THE TOTAL LENGTH OF THE PIECE BASED ON TIME SIGNATURE.
- EACH MATH CONCEPT GENERATES A NUMBER SERIES THAT IS THEN CONVERTED TO MUSICAL NOTES
- INDIVIDUAL TRACKS ARE CREATED FOR EACH INSTRUMENT WITH SPECIFIC PROGRAM, OCTAVE AND MIDI CHANNEL.
- NOTES ARE MAPPED TO SCALES AND APPENDED TO THEIR RESPECTIVE TRACKS.

MUSIC ANALYSIS LOGIC



- THE SYSTEM FIRST IDENTIFIES THE INSTRUMENT TYPE
- THEN EACH TRACK IS EVALUATED BASED ON 4 PRIMARY METRICS:
  - STEP: INTERVAL SIZE
  - RHYTHM: SPEED AND COMPLEXITY OF PATTERN
  - HARMONY: RELATIONSHIP BETWEEN NOTES
  - CONTOUR: THE SHAPE OF THE MELODY
- A POINT-PENALTY METHOD IS APPLIED TO THE METRICS WITH +2 FOR MEETING THE CRITERIA AND -1 FOR FAILING.
- AN INSTRUMENT MUST ACHIEVE AT LEAST 3 OUT OF 4 METRICS TO BE LABELED AS "GOOD FIT"

RESULTS

Instrument Analysis Results

No.	Instrument	Math Concept Used	Interval Distribution	Rhythm Density	Harmonic Complexity	Melodic Contour
1	Violin	Fibonacci Sequence	The Violin has small step sizes, giving smooth melodic movement.	The Violin has low rhythmic density.	The Violin has rich harmonic complexity.	The Violin generally ascends.
2	Trombone	Prime Numbers	The Trombone has large step sizes, creating dramatic melodic leaps.	The Trombone has low rhythmic density.	The Trombone has moderately complex harmony.	The Trombone generally ascends.
3	Clarinet	Multiples of 2	The Clarinet has small step sizes, giving smooth melodic movement.	The Clarinet has low rhythmic density.	The Clarinet has rich harmonic complexity.	The Clarinet generally ascends.
4	Snare Drum	Multiples of 5	The Snare Drum has small step sizes, giving smooth melodic movement.	The Snare Drum has low rhythmic density.	The Snare Drum has simple harmonic structures.	The Snare Drum has a balanced melodic contour.

Does the selected mathematical concept work well for the instrument?

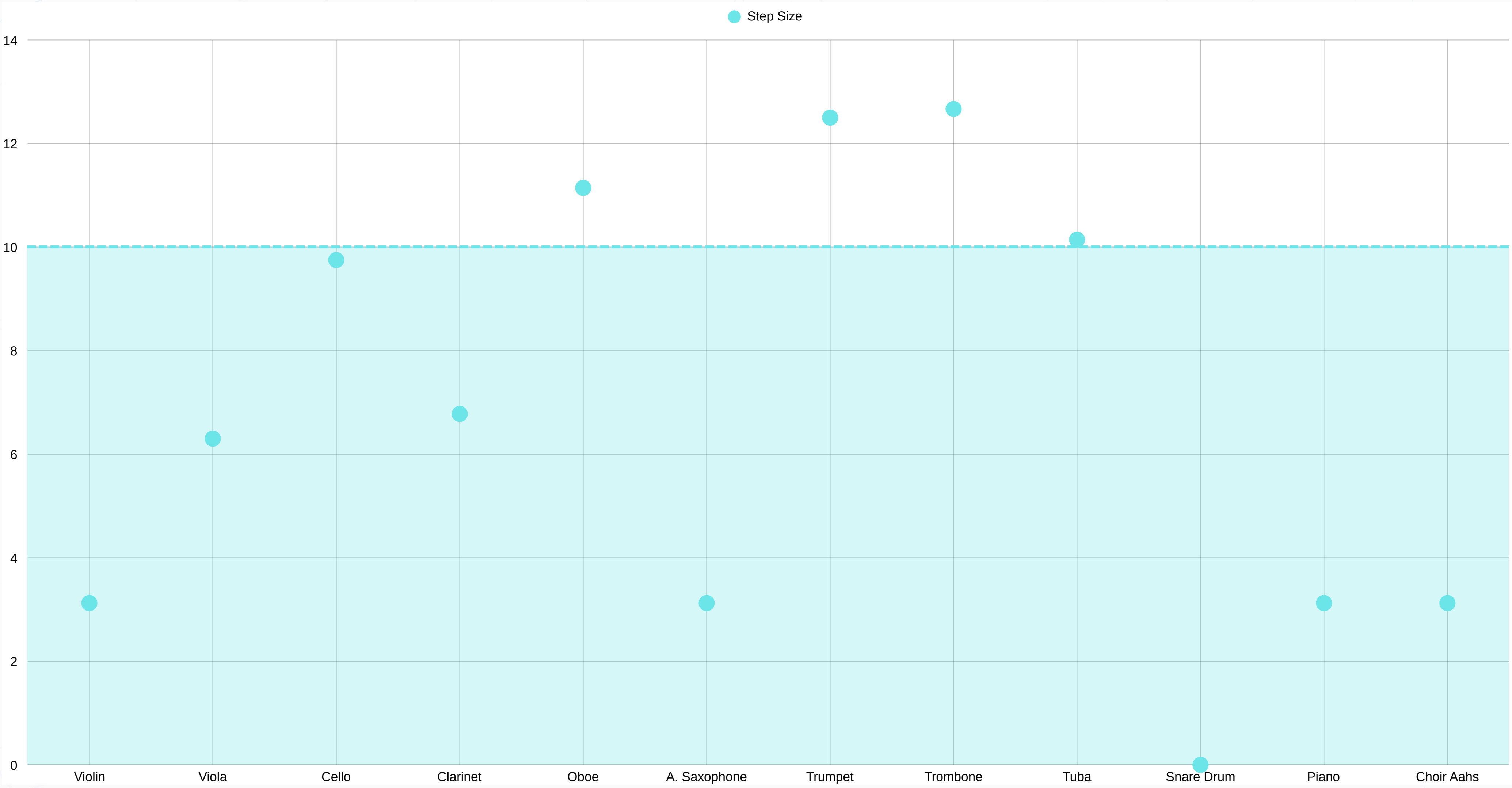
Violin: Yes - The step size fits well with string instruments. The rhythm density fits well with string instruments. Consider using simpler harmonic structures for string instruments. The melodic contour fits well with string instruments.

Trombone: No - Consider using larger step sizes for brass instruments. The rhythm density fits well with brass instruments. Consider using low harmonic complexity for brass instruments. The melodic contour fits well with brass instruments.

Clarinet: No - The step size fits well with woodwind instruments. Consider using moderate rhythm density for woodwind instruments. Consider using moderate harmonic complexity for woodwind instruments. The melodic contour fits well with woodwind instruments.

Snare Drum: Yes - Step size is less relevant for percussion instruments. Consider using moderate rhythm density for percussion instruments. The harmonic complexity fits well with percussion instruments. Melodic contour is less relevant for percussion instruments.

INTERVAL DISTRIBUTION FOR PRIME NUMBERS



CONCLUSION

WE CONCLUDE THAT MATHEMATICAL CONCEPTS CAN DEFINITELY BE USED TO GENERATE MUSIC. HOWEVER NOT EVERY MATH CONCEPT WILL WORK FOR EVERY INSTRUMENT. THIS IS BECAUSE EACH INSTRUMENT HAVE DIFFERENT MUSICAL CHARACTERISTICS AND NOT EVERY MATHEMATICAL PATTERN TRANSLATES WELL WITH THEM.

- STRING INSTRUMENTS HANDLE IRREGULAR PATTERNS LIKE PRIME AND FIBONACCI MORE NATURALLY BECAUSE THEIR PITCH CHANGES ARE SMOOTH AND CONTINUOUS.
- BRASS AND PERCUSSION INSTRUMENTS RESPOND BETTER TO STABLE REPEATING PATTERNS OF MULTIPLES OF 2S AND 5S.
- WOODWINDS CAN PLAY ALL 4 MATH CONCEPTS FAIRLY WELL AS THEY ARE FLEXIBLE ENOUGH FOR UNEVEN PATTERNS AND SMOOTH WITH REGULAR PATTERNS.

FUTURE ENHANCEMENTS

IN THE FUTURE, WE WOULD LOVE TO INCLUDE MORE COMPLEX MATHEMATICAL CONCEPTS SUCH AS SET THEORY, RATIOS, AND FRACTALS. IN ADDITION, WE WOULD ALSO LOVE TO TRANSFORM THIS PROJECT INTO A MOBILE APP THAT CAN BE AVAILABLE ON APP STORES.

REFERENCES

STANDARD MIDI FILES - HTTPS://MIDO.READTHEDOCS.IO/EN/STABLE/FILES/MIDI.HTML

MUSIC21 - A TOOLKIT FOR COMPUTER-AIDED MUSICOLOGY

HTTPS://MUSIC21.ORG/MUSIC21DOCS/

STREAMLIT API REFERENCE HTTPS://DOCS.STREAMLIT.IO/DEVELOP/API-REFERENCE

ALGORITHMIC MUSIC COMPOSITION WITH PYTHON

HTTPS://WWW.YOUTUBE.COM/WATCH?V=37S0ZDPJPC4

ALGORITHMIC COMPOSITION: THE MUSIC OF MATHEMATICS BY CARLO ANSELMO

HTTPS://WWW7.HSC.EDU/DOCUMENTS/ACADEMICS/MATHCS/PENDERGRASS/ALGORITHMICCOMPOSITIONPAPER_HSCHONORS.PDF?HL=EN-US

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